Assignment 1

CSE 423

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Section : 10

Category A:

(1) Given,

(120, 23) and (423, 428)

$dx = (423 - 120) = 303$

$dy = (428 - 23) = 405$

Since $|dy| > |dx|$, steps $= 405$

$\therefore$ The number of times x will be increased is 303 times

Total pixels needed $= (\text{Steps} + 1) = (405 + 1) = 406$

(2) In DDA algorithm,

when the slope is between -1 and 1, the line is more hizontal. Here, for every step in x, y changes by 1 unit. So incrementing x by 1, computing y from the line equation gives a pixel in every column producing a connected line. Meanwhile, when slope is less than -1 and greater than 1, the line is more vertical. Here, for every step in x, y changes by more than 1 unit. So incrementing x by 1, we would skip multiple values of y leaving gaps in the line. Thus we increment y by 1 and compute x from the line equation. This guarantees a pixel in every row, keeping the line unbroken.

(4) Yes Midpoint algorithm is better than DDA.

In DDA algorithm, it is slow because it uses decimals, forcing the computer to constantly do heavy math and then round those decimals to nearest whole pixel at every step which is inaccurate because these decimal rounding errors causes the line to drift away from original destination.

We solve this by using Midpoint Algorithm.

It uses clever math. to stick strictly to whole numbers. It completely removes rounding process and ensures 100% accurate lines.

Category B

(8)(a) Given,

(5,16) to (13,10)

$m = \frac{10-16}{13-5} = \frac{-3}{4} = -0.75$

Since $-1<m<1$

So, $x = x+1$

$y = y-|m|$

x (+1)	y(-0.75)	y(round)	pixels
5	16	16	(5,16)
6	15.25	15	(6,15)
7	14.25	14	(7,14)
8	13.75	14	(8,14)
9	13	13	(9,13)
10	12.25	12	(10,12)
11	11.5	12	(11,12)
12	10.75	11	(12,11)
13	10	10	(13,10)

(b) (0,0) to (4,6)

$$m = \frac{6-0}{4-0} = 1.5$$

Since $-1 < m$, $y = y+1$

$x = x + \frac{1}{|m|}$

$y(+1)$	$x(+\frac{1}{1.5})$	x(rounded)	pixels
0	0	0	(0,0)
1	0.67	1	(1,1)
2	1.34	1	(1,2)
3	2.01	2	(2,3)
4	2.68	3	(3,4)
5	3.35	3	(3,5)
6	4.02	4	(4,6)

10) Given,

$$\frac{x}{7} - \frac{y}{12} = 5$$

If $y = 12 \Rightarrow \frac{x}{7} - 1 = 5$

$\Rightarrow \frac{x}{7} = 6$

$\therefore x = 42$

Starting point $= (42, 12)$

If $y = 0 \Rightarrow \frac{x}{7} - 0 = 5$

$x = 35$

Ending point $= (35, 0)$

$dx = 35 - 42 = -7$

$dy = 0 - 12 = -12$

$|dy| > |dx|$

$\therefore$ zone $= 5$

Now, $(42,12) = (-12,-42)$

$(35,0) = (0,-35)$

$dx' = 0+12 = 12$

$dy' = -35+42 = 7$

$D = 2dy' - dx' = 2(7)-12 = 2$

$\Delta NE = 2(dy' - dx') = 2(7-12) = 2(-5) = -10$

$\Delta E = 2dy' = (2\times 7) = 14$

x	y	d	E/NE	d (update)	x'	y'	Pixels
-12	-42	2	NE	2(-10)=-8	42	12	(42,12)
-11	-41	-8	E	-8+(14)=6	41	11	(41,11)
-10	-41	6	NE	6+(-10)=-4	41	10	(41,10)
-9	-40	-4	E	-4+14=10	40	9	(40,9)
-8	-40	10	NE	10+(-10)=0	40	8	(40,8)
-7	-39	0	E	0+14=14	39	7	(39,7)